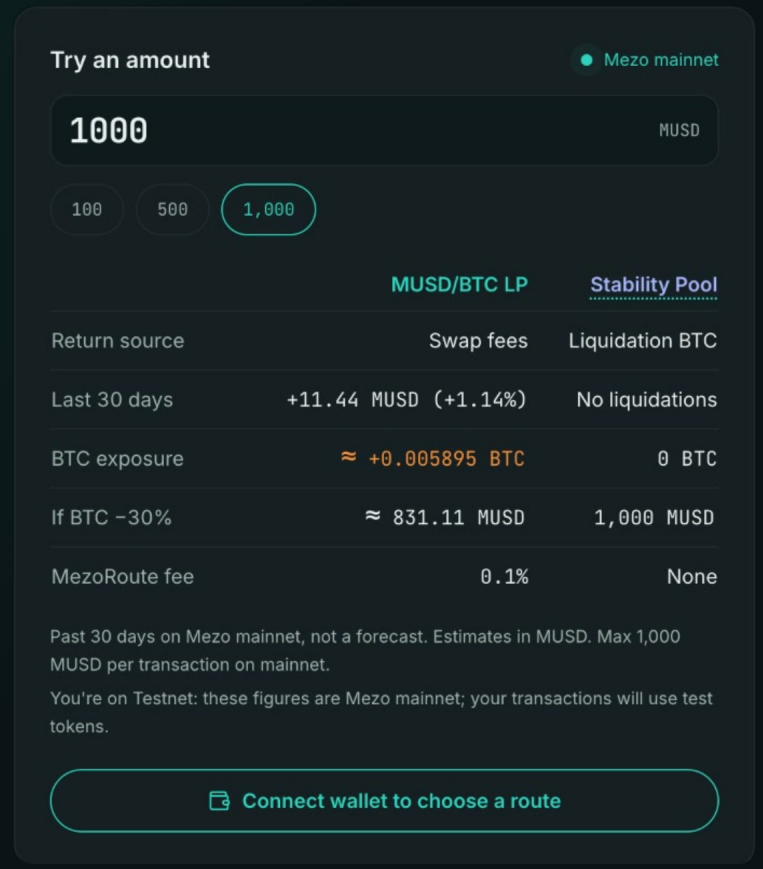
MEZOROUTE · BUILD WITH MUSD AND MEZO · WAVE 1

Pick a route.
Know the BTC risk.

MezoRoute shows Mezo borrowers where to put their MUSD (the MUSD/BTC pool or the Stability Pool) by what it does to their BTC risk, and runs borrow → LP in one transaction.

Live on Mezo mainnet (capped) and testnet



mezoroute.vercel.app · github.com/hms1499/MezoRoute

THE PROBLEM

Borrowed MUSD has two homes and no map

01

Which route?

The MUSD/BTC pool and the Stability Pool earn in different ways and fail in different ways. Nothing puts them side by side.

02

Too many steps

Borrow, swap and add liquidity are separate screens, approvals and transactions. The swap size is a guess.

03

Hidden BTC risk

Borrowing against BTC, then holding half the LP in BTC, stacks BTC exposure. A BTC drop hits the Trove and the LP at once.

MezoRoute answers all three on one screen: compare, preview, execute.

Persona: a Mezo borrower who holds, or can borrow, MUSD against BTC

MEASURED IN THE MEZO APP

Borrow → LP today: 6 wallet confirmations

Screen	Wallet confirmation	Count
Borrow	Borrow more MUSD	1
Swap	Approve MUSD, swap to BTC	2
Pool	Approve MUSD, approve BTC, deposit	3
Total	3 screens	6

MEZOROUTE BORROW & DEPLOY

1 tx

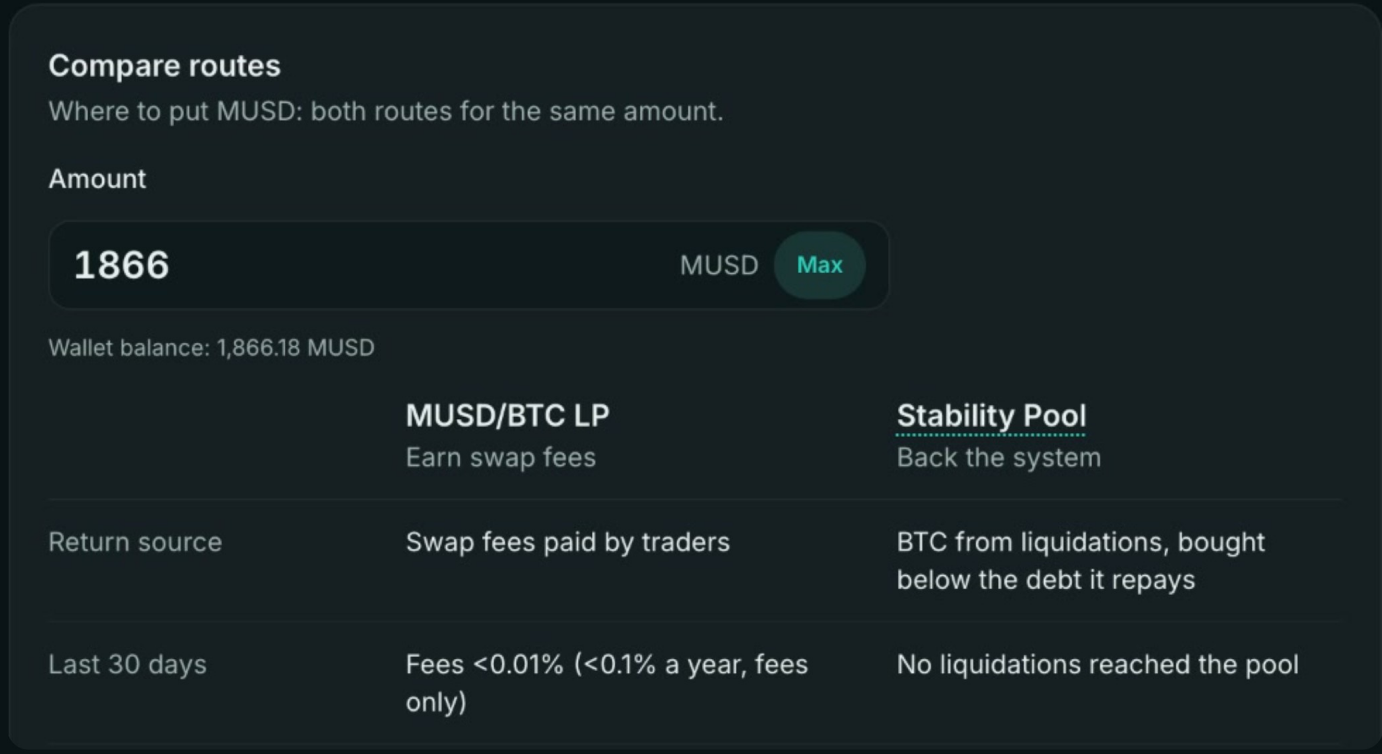
plus 2 gas-free signatures, on one screen. The swap size is computed exactly; leftovers return to the wallet.

Measured 19 Oct 2026 on testnet.mezo.org with 10 MUSD: 6 transactions sent, all succeeded. Exact approvals, so every new deposit repeats them. Sign-in excluded on both sides.

COMPARE ROUTES

Two routes, one BTC-risk view

- Return source: swap fees or liquidation BTC
- Last 30 days from on-chain history, not a forecast
- BTC exposure next to the user's Trove
- "If BTC moves" from −30% to +30%

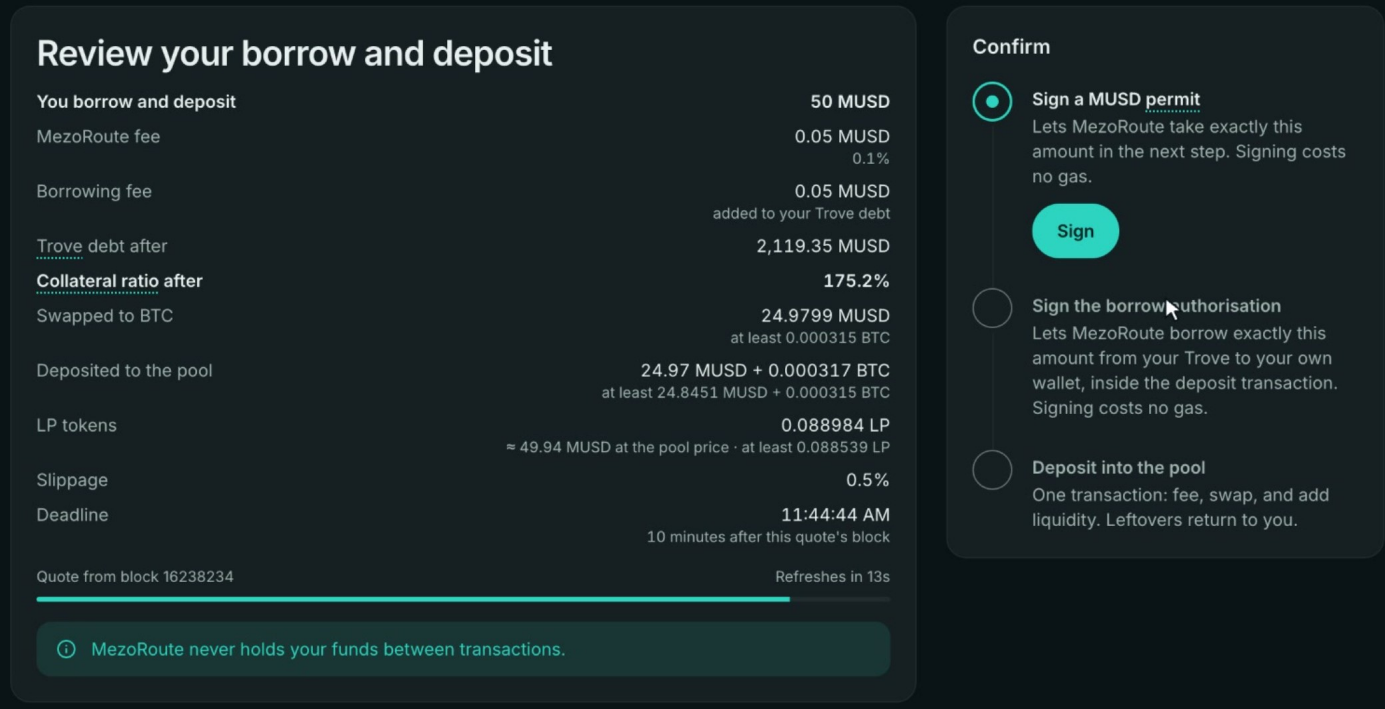


BORROW & DEPLOY

Borrow → LP in one transaction

1. Sign a MUSD permit
2. Sign the borrow authorisation (EIP-712, Mezo's BorrowerOperationsSignatures)
3. One transaction: borrow, swap the exact share, add liquidity, return leftovers

Minimum outputs, a deadline and a 160% collateral-ratio floor are checked before signing.



MEZO INTEGRATION

Three MUSD surfaces, used in depth

Borrowing

BorrowerOperationsSignatures: a contract borrows for the user with one EIP-712 signature. Trove hints, fee and recovery mode read live.

MUSD/BTC pool

Tigris pool and router: exact optimal swap, permit entry and exit, fee claims, cost basis from events.

Stability Pool

Deposit, withdraw and claim BTC gains; Liquity P/S maths for a 30-day backtest; paused withdrawals handled.

> **Why not MEZO?**
>
> No gauge pays MEZO, veMEZO is not deployed, and mainnet MUSD/BTC pools have no gauge. We chose a deep MUSD integration over a surface MEZO one.

VALUE TO MEZO

More MUSD minted, a deeper MUSD/BTC pool

29.7M
MUSD supply on mainnet

48%
of it sits in the Stability Pool (14.2M MUSD)

59.2k
MUSD in the basic MUSD/BTC pool, with 0.71 BTC

Borrow & Deploy turns Trove headroom into new MUSD and new pool depth. Compare routes sends idle MUSD to Mezo's own destinations, not off-chain.

Source: Mezo mainnet contract reads at block 12,618,417, 19 Oct 2026

TECHNICAL IMPLEMENTATION

A narrow executor, tested to the wei

Contract

- No owner, admin or proxy; immutable 1,000 MUSD cap per entry
- Payouts are balance deltas inside the call; allowances reset to 0
- Funds never rest in the contract between transactions
- 75 Foundry tests: fuzz, invariants, reentrancy, signature misuse

App

- Quote engine mirrors pool rounding: every live entry replays exactly (14/14 testnet, 1/1 mainnet)
- Results from events, never wallet balance changes
- Every error mapped to a message and a recovery action
- 806 unit tests; static Next.js export

Solidity 0.8.24 · OpenZeppelin 4.9 · Tigris · Next.js 14 · wagmi · viem · Mezo Passport · self-audit in contracts/SELF_AUDIT.md

PROJECT READINESS: MAINNET DEPLOYMENT

Real round trips on Mezo mainnet

Action	Result	Tx
Executor deployed	Source verified, 10 bps fee, 1,000 MUSD cap	0x8aa5…b1d
Stability Pool	Deposit 10 MUSD, then withdraw	0x060d…7fe
LP enter	10 MUSD → 0.017446 LP, quote exact (0 bps)	0xb401…0fb
Claim fees	Pool fees claimed from Positions	0xde86…521
LP exit (Max)	9.9636 MUSD back; executor holds nothing	0x48c1…147

0.0264
MUSD total cost of a 10 MUSD LP round trip

0 bps
quote error on the mainnet entry

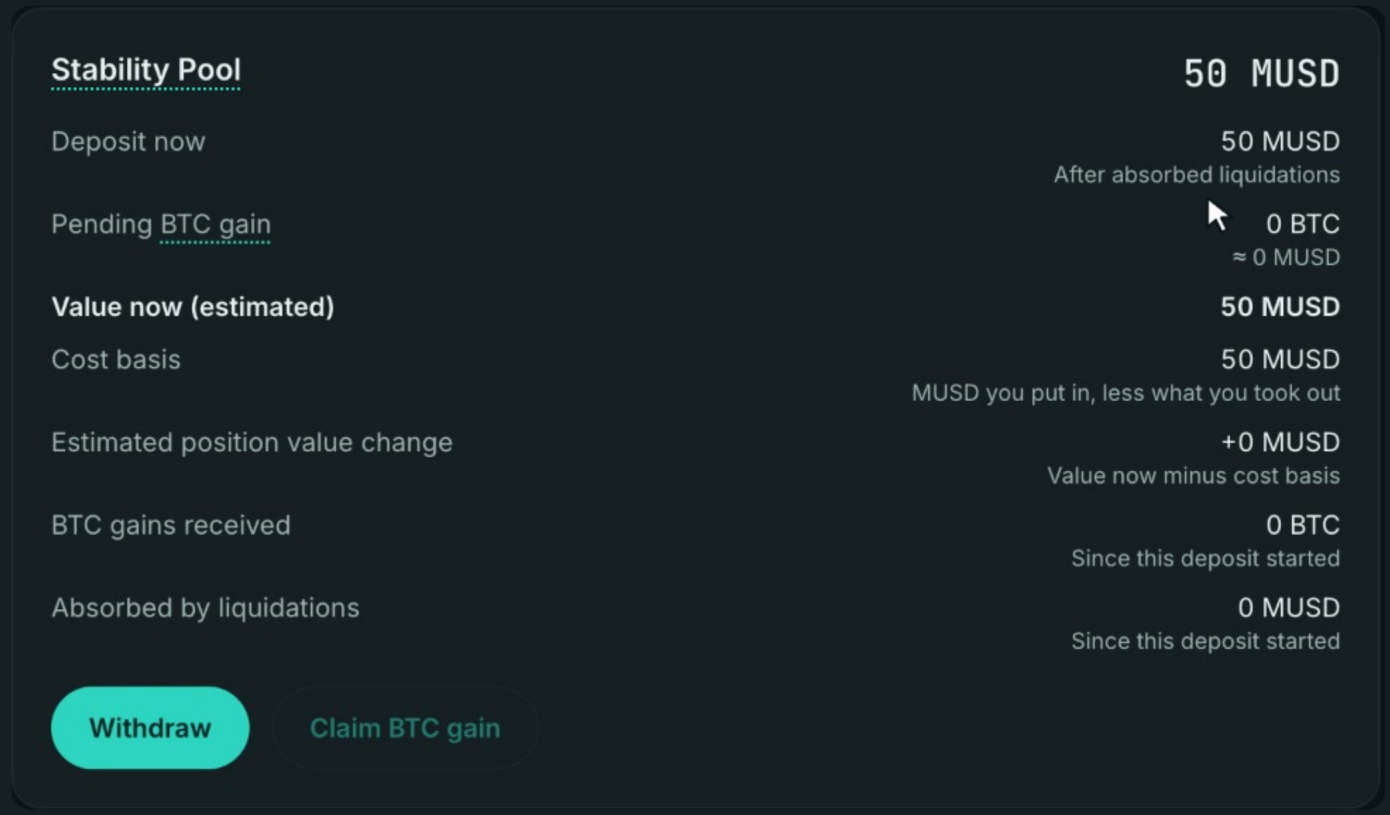
0
tokens left in the executor afterwards

Executor 0x046449b1229C863e278f369869BA10e12bd80d9E · 15 Oct 2026 · full hashes in the README

USER EXPERIENCE

Bitcoin wallets, honest results

- Unisat, OKX and Xverse through Mezo Passport, plus EVM wallets
- Positions: value, cost basis, claimable fees, BTC gains
- Mezo terms explained in place; works on phones from 360 px wide
- Primary action within three screens of connecting



BUSINESS MODEL AND USERS

One disclosed fee, charged only on entry

0.1%

- On MUSD entering the LP route; live on mainnet now
- Fixed at deploy, hard-capped at 0.5% in the contract
- Shown before signing and emitted on chain
- No exit fee, no Stability Pool fee

Who it is for

- Mezo borrowers who hold, or can borrow, MUSD against BTC
- Know the DeFi basics; don’t want to compute pool ratios or Trove hints
- EVM or Bitcoin wallet, desktop or mobile
- Track 2 (access and distribution) first, Track 1 (DeFi) second

NEXT MILESTONES

From one route to a MUSD toolkit

Wave 2 · by 15 Nov 2026

BTC leverage and unwind

One transaction each, using pool flash swaps and Mezo’s signed Trove adjustment; testnet and capped mainnet.

After the hackathon

More routes

New routes as separate executors, and opening a Trove in the same flow.

After the hackathon

Audit, then scale

Smart-account support and an external audit before raising the caps.

> Shipped in Wave 1: both routes, Borrow & Deploy, positions with fee claims, Bitcoin wallets, live on mainnet.

Pick a route.
Know the BTC risk.

TRY IT

mezoroute.vercel.app

github.com/hms1499/MezoRoute

Unaudited software with an immutable 1,000 MUSD cap per entry on mainnet. Testnet tokens have no value.